Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Development of Dual Cavity Package and Module for High-Reliability Sensors*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Electrical – Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	ICT Modules and Components	*Mobile Communication Modules and Components*	*Electrical and electronic components*	*Sensor components*
Concept of Proposed Technology	*The proposed technology is a dual cavity type package and module platform for sensor integration, designed to mount optical sensors, acceleration sensors, and other multi-functional sensor ICs in a single compact module.* *As communication modules as well as aerospace and defense components increasingly require a reduced form factor and higher functional integration, this project proposes a multi-sensor integration technology that enables multiple sensor ICs to be assembled within a single package.* *To achieve this, the proposed package is designed with a flip-type multi-cavity structure, in which multiple IC mounting cavities are formed within a single ceramic package, and the assembled ICs are electrically interconnected and operated inside the package.*			

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	The proposed technology is based on a ceramic hermetic package platform consisting of a ceramic body, metal lid sealing, and a getter lid, and requires advanced ceramic material and process technologies such as high-resolution ceramic paste and green sheet tape processes. *Based on these core technologies, the project aims to develop a high-reliability, hermetic dual cavity type sensor package and to further realize a compact, high-integration sensor module by assembling sensor devices into the package.*
Rationale for Joint R&D Project	*The development of a dual cavity, multi-chip integrated ceramic package for high-reliability sensor modules requires advanced HTCC (High Temperature Co-fired Ceramic) [1,2] material and manufacturing technologies, including alumina tape, tungsten conductor paste formulation, multilayer structuring, co-firing process optimization, and high-precision dimensional control.* *Fraunhofer IKTS possesses world-class core technologies in HTCC materials, multilayer ceramic processing, and high-reliability ceramic packaging, which are essential for realizing fine-pitch, multi-cavity ceramic structures and stable manufacturing processes.* *Through this joint R&D project, a technical collaboration from Fraunhofer IKTS and the Korean partners will be pursued, enabling the establishment of an independent domestic manufacturing and design capability for advanced ceramic packages.* *In addition, Fraunhofer IKTS maintains strong cooperative networks with German and European ceramic and sensor-related companies, which provides a strategic pathway to connect the project outcomes to commercialization and business expansion with local German industrial partners.* *While the Korean industrial partner will focus on package structure design, module integration, reliability evaluation, and commercialization, the German partner (Fraunhofer IKTS) will contribute its original HTCC material and process technologies and process optimization know-how.* *Through this international joint R&D project, the partners aim to significantly reduce technical risks, accelerate development timelines, and secure global competitiveness in advanced sensor packaging technologies for aerospace, defense, and high-reliability industrial applications.*
Key Words	*1. Dual Cavity Ceramic Package* *2. Fine HTCC (High Temperature Co-fired Ceramic)* *3. Multi-Sensor Integration Module* *4. Hermetic Sealing* *5. Getter Integrated Packaging*

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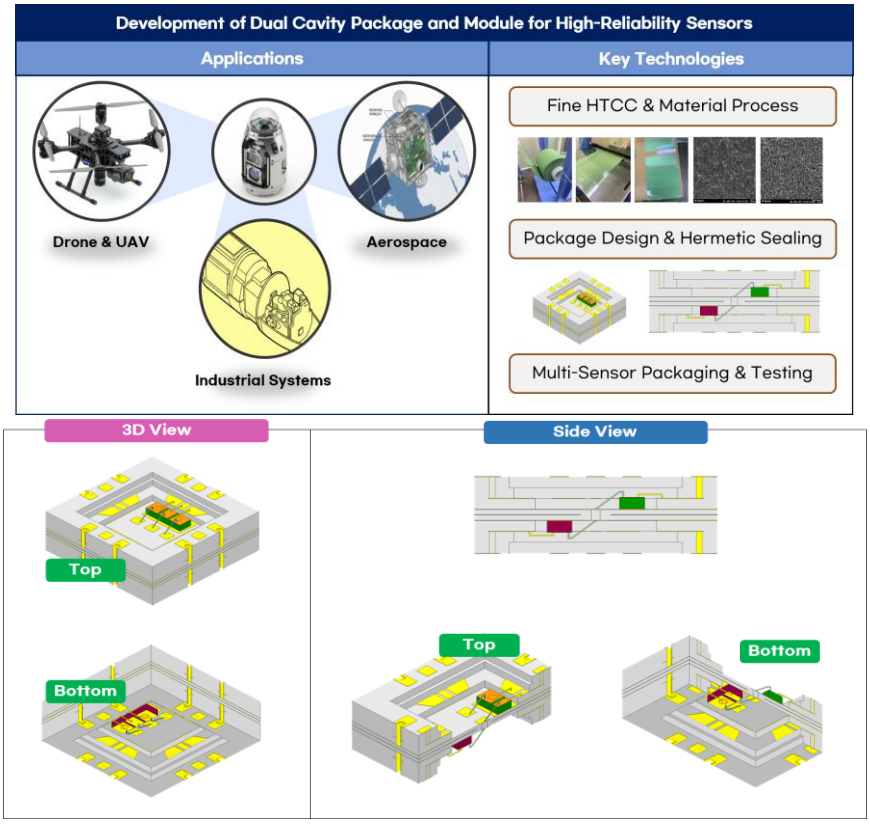
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (8)
R&D Objectives	*The ultimate objective of this project is to develop a high-reliability HTCC-based ceramic package and multi-sensor module platform with a dual cavity structure for sensors (e.g., inertial, acceleration sensor) applications, as illustrated in Figure 1, which presents the application domains, key technologies, and the structural concept of the dual-cavity HTCC package, including its top view and side view configurations for multi-sensor integration.* *To achieve this goal, the following specific technical objectives are defined:* *(1) Development of an HTCC ceramic package structure with dual mounting cavities for multi-sensor/IC integration.* *(2) Establishment of advanced HTCC manufacturing process technologies, including multilayer ceramic structure design, co-firing process optimization, fine-pitch cavity formation, and high-precision dimensional control.* *(3) Development of hermetic sealing technologies, including metal lid sealing and getter lid technologies in lid-attached or lid-integrated configurations.* *(4) Development of internal interconnection/routing and integration technologies, enabling multiple sensor ICs to be assembled within a single package and electrically interconnected to operate as a module.* *(5) Realization and verification of a compact, high-reliability multi-sensor module through prototype fabrication and reliability evaluation.* 			

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R&D Target Specifications

Figure 1. Concept and Structural Design of a Dual-Cavity HTCC Package and Multi-Sensor Module for High-Reliability Sensors, Including Top View and Side View.

The target performance specifications of the dual-cavity HTCC-based ceramic package and multi-sensor module to be developed in this project are as follows:

Category	Item	Target Specification
Hermeticity	He Leak Rate	≤ 1 × 10^{-8} atm·cc/sec (Fine leak)
	Reliability	1,000 hours at 125°C
PKG Structure	Number of Cavities	≥ 2
	Number of ICs per Cavity	At least 1 per cavity
HTCC Process	Dimensional Variation (after Co-firing)	±0.3%
	Line Width	≤100um (≤ 80um)
	Line Pitch	≤100um
	Via Diameter	≤100um
Reliability	Thermal Cycling	-55°C / +125°C, ≥ 500 cycles, no failure
	Life Test	85°C / 85% RH, ≥ 1,000 hours, no failure
	Mechanical Test	Pass (Vibration & Shock test @Mil Std.-883 or equivalent)
Electrical Performance	Internal Interconnection Resistance	≤ 100 mΩ per channel

R&D Plan

1. Overall R&D Strategy

This project aims to develop an HTCC-based dual-cavity package and a multi-sensor module platform. The development will be carried out using a structured five-gate strategy (WP1 to WP5), progressing as follows:

Structure design → Process development → Hermetic sealing → Module integration → Reliability verification

This phased approach ensures systematic advancement from fundamental package architecture to final system-level validation. In particular, the project will be implemented under a role-sharing collaborative framework between the Korean partner and Fraunhofer IKTS:

- ***Korean partner:*** *Responsible for package structure design, development of hermetic sealing technologies, module integration, and overall system integration.*
- ***Fraunhofer IKTS:*** *Responsible for HTCC material and process technologies, HTCC substrate fabrication, and manufacturing process optimization.*
- ***Korean partner:*** *Responsible for defining sensor module requirements, module assembly, and application-level validation.*

The project will be executed through a five- stage-gate approach to develop an HTCC-based dual-cavity multi-sensor package and module as follows:

- *WP 1 – Structure Design (Months 0 - 6): The Korean partner designs the dual-cavity package architecture and interfaces. The Korean partner also defines sensor requirements and application constraints. Fraunhofer IKTS advises on HTCC material systems and manufacturability.*
- *WP 2 – Process Development (Months 0-15): Fraunhofer IKTS leads HTCC material selection, substrate fabrication, and process optimization. The Korean partner validates process compatibility with the package*

design and confirms assembly feasibility.

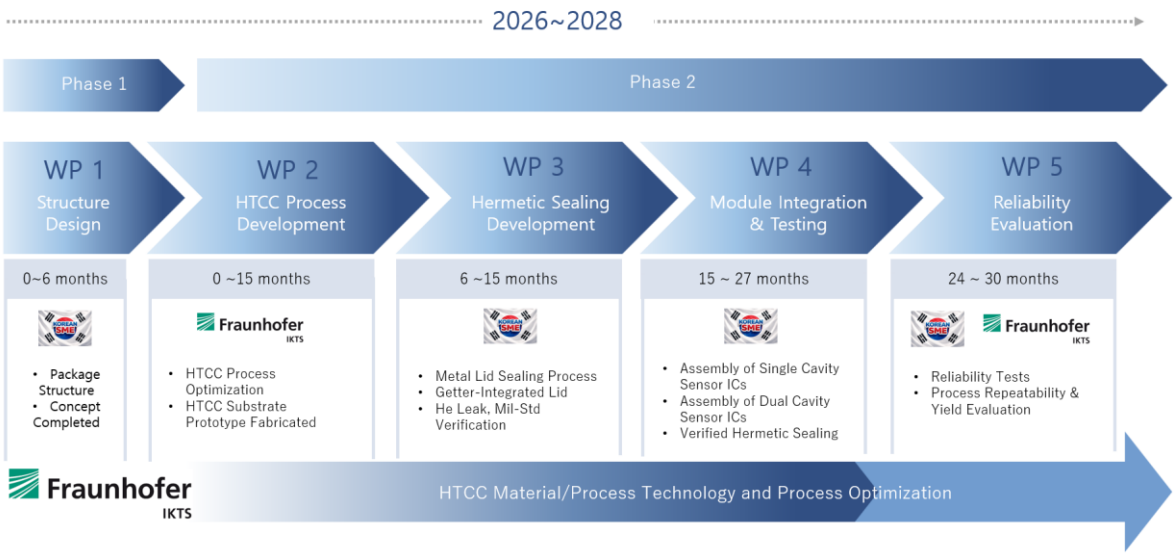
- *WP 3 – Hermetic Sealing (Months 6-15): The Korean partner develops and validates hermetic sealing for the dual cavities, with Fraunhofer IKTS supporting material and interface optimization. The Korean partner also verifies sensor compatibility.*
- *WP 4 – Module Integration (Months 15–27): The Korean partner performs package and system integration. The Korean partner leads sensor assembly, calibration, and functional validation. Fraunhofer IKTS supports substrate and assembly compatibility.*
- *WP 5 – Reliability Verification (Months 24–30): The Korean partner conducts environmental and lifetime reliability testing, with Fraunhofer IKTS supporting failure analysis and HTCC material stability assessment.*

Key Methodologies include:

- *HTCC material characterization and process optimization*
- *CAD/CAE design and thermo-mechanical simulation*
- *Hermetic sealing development and helium leak testing*
- *Sensor assembly, calibration, and functional testing*
- *Reliability testing (thermal cycling, humidity, mechanical stress)*

Expected Milestones

1. *HTCC Dual-Cavity Package and Process Readiness: The dual-cavity package structure is finalized and a stable HTCC substrate fabrication process is established, demonstrating manufacturability [M12].*
2. *Validated Hermetic Dual-Cavity Package: Reliable hermetic sealing for both cavities is achieved and verified, meeting leak-tightness and mechanical integrity requirements [M20].*
3. *Reliability-Qualified Multi-Sensor Module: A fully integrated HTCC-based dual-cavity multi-sensor module successfully passes functional and reliability testing, confirming readiness for application and scale-up [M30].*



2. Continuous Contribution of Fraunhofer IKTS

Throughout the entire project period, Fraunhofer IKTS will continuously support HTCC material and process technologies and process optimization, as indicated in the lower part of the roadmap:

"HTCC Material / Process Technology and Process Optimization."

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*4.1. Roles and Responsibilities of the Research Institute (Fraunhofer IKTS)* *Fraunhofer IKTS will act as the core research institute in the field of HTCC materials and process technologies in this project and will lead the establishment of fundamental manufacturing technologies for realizing the dual-cavity ceramic package.* *(1) Technical Contributions* ● *Development and optimization of ceramic pastes and tapes for HTCC* ● *Development of multilayer ceramic lamination, cavity formation, and via/routing technologies* ● *Optimization and stabilization of co-firing process conditions* ● *Establishment of high-precision dimensional control and control of shrinkage and warpage* *(2) Research Scope and Tasks* ● *Leading HTCC process development and manufacturability verification, mainly in Phase 2* ● *Fabrication of test coupons and HTCC substrate prototypes* ● *Derivation of process windows and definition of process limit conditions* ● *Verification of HTCC process feasibility for the package structures designed by Korean partner* *(3) Resources and Infrastructure to be Provided* ● *HTCC material development and processing facilities (mixing, forming, lamination, firing, etc.)* ● *Dedicated expert researchers in ceramic process technologies* ● *Analysis and characterization equipment (dimensional inspection, microstructure, and material property evaluation)* *(4) Collaboration and Joint Research Role* ● *Co-design collaboration with Korean partner between process and package structure* ● *Provision of design guidelines reflecting process constraints* ● *Continuous support of HTCC material/process technologies and process optimization throughout the project period*
Expected R&R of Korean SME Partner	*4.2. Roles and Responsibilities of the Korean SME Partner* *Korean Partner, as the project coordinator, will be responsible for the entire development cycle from package structure design to module integration, reliability verification, and commercialization.* *(1) Technical Contributions* *Structural design of the dual-cavity ceramic package* *Design of internal routing structures, lid structures, and getter application schemes* ● *Development of hermetic sealing processes and package sealing technologies* ● *Development of sensor assembly and module integration technologies* *(2) Development Tasks* ● *Overall leadership of Phases 1, 3, 4, and 5* ● *Development of metal lid sealing and getter application processes* ● *Assembly and functional verification of single-cavity and dual-cavity sensor packages* ● *Fabrication and evaluation of module prototypes* *(3) Testing and Validation Activities* ● *Helium leak tests and hermeticity evaluation* ● *Reliability tests such as temperature cycling, high-temperature & humidity, vibration, and shock tests*

- *Evaluation of process repeatability and yield*

(4) Resource Commitment and Commercialization Role

- *Deployment of engineers for package design and process development*
- *Provision of package/module manufacturing infrastructure and measurement equipment*
- *System-level applicability verification in collaboration with end-users (another Korean partner)*
- *Leadership in mass production and commercialization after project completion*

4.3. Roles and Responsibilities of the End-user / Demonstration Partner (Korean partner)

Another Korean partner will participate in this project as a sensor module specialist and end-user company, and will be responsible for verifying the applicability of the developed technology to real systems and linking the project outcomes to commercialization.

(1) Technical Contributions

- *Providing practical requirements of target sensor devices (such as optical sensors and acceleration sensors), including specifications, interfaces, and reliability requirements*
- *Providing design feedback from a sensor module perspective on package structure, electrical interfaces, and assembly methods*
- *Defining system-level requirements such as miniaturization, reliability, and environmental robustness*

(2) Roles and Participation in Development

- *Participation in technical reviews and joint verification activities mainly in Phases 3, 4, and 5*
- *Supplying their own sensor devices and cooperating in sensor assembly and module integration with project leader (Korean partner)*
- *Performing demonstration tests and application-level verification for single-cavity and dual-cavity sensor module configurations*

(3) Testing and Validation Activities

- *Conducting functional tests, performance verification, and environmental reliability evaluation from a sensor module perspective*
- *Providing validation scenarios reflecting real customer requirements*
- *Identifying design improvement points from the viewpoint of mass production and practical deployment*

(4) Commercialization and Dissemination Role

- *Acting as the lead customer by applying the project results to their own sensor module products*
- *Establishing initial mass-production and market introduction strategies in cooperation with project leader (Korean partner)*
- *Contributing to market dissemination and reference creation using their domestic and global customer networks*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*By achieving the proposed quantitative target specifications such as hermeticity, miniaturization, integration density, and reliability, this project aims to generate the following technical, business, and industrial impacts.* *1. Technical and Scientific Impact* *This project will create a significant technical achievement by realizing advanced dual-cavity HTCC-based ceramic package structures, high-precision multilayer ceramic processes, and hermetic sealing technologies, which are still rarely commercialized domestically. Through this project, the following technical outcomes will be achieved:* ● *Demonstration of dual-cavity multi-sensor integrated package structures* ● *Establishment of high-precision HTCC process technologies and design rules* ● *Securing hermetic sealing technologies based on metal lid and getter integration* ● *Acquisition of core technologies for high-reliability sensor packaging* *These achievements will serve as a technological foundation to elevate the domestic ceramic packaging technology level to that of global leading groups, and the developed technology can be utilized as a platform technology expandable to various sensor and semiconductor packaging applications.* *2. Business Impact* *The developed technology can be utilized as a high-value-added sensor module packaging platform for aerospace, defense, optical communication, and high-reliability industrial systems. By providing a high-integration and compact solution compared to conventional single-sensor packages, this project will enable:* ● *Entry into premium markets* ● *Creation of new high-value product lines* ● *Securing new revenue streams for Korean and participating partners* *In addition, through joint R&D with Fraunhofer IKTS, commercialization opportunities linked to German and European markets will be established.* *By leveraging End-user / Demonstration Partner (Korean partner) as a lead customer, early market validation and reference creation will be achieved, forming a foundation for global customer acquisition and export expansion.* *3. Industrial and Economic Impact* *This project is expected to generate the following broader industrial and economic impacts:* *Advancement and localization of high-end ceramic packaging technologies in Korea* *Strengthening the domestic supply chain competitiveness in aerospace, defense, and high-reliability sensor components. Import substitution effects in high-reliability packaging fields with strong overseas dependency* ● *Enhancement of global technology cooperation networks and national technological reputation through international joint R&D* *Furthermore, the technologies secured through this project are expected to play a catalytic role in upgrading the overall technological level of the domestic ceramic packaging ecosystem.*

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6. References

(1) R. R. Tummala, E. J. Rymaszewski, and A. G. Klopfenstein, Eds., *Microelectronics Packaging Handbook: Semiconductor Packaging*, 2nd ed. New York, NY, USA: Chapman & Hall, 1997.

(2) A. Elshabini and F. D. Barlow III, *Ceramic Interconnect Technology*. Boston, MA, USA: McGraw-Hill Education, 1998.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization